

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 11

1

Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!

2



ALVARO: Start the recording!



3

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Double Integrals over Regions

4

Today – Double Integrals in Regions!

- General Regions
- Regions of Type I and II
- Changing the Order of Integration
- Properties of Double Integrals

5

Regions of Type I and II

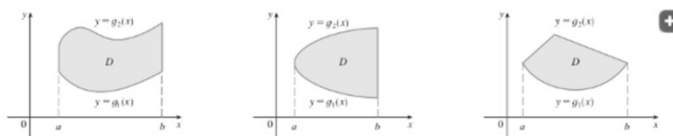
A plane region D is said to be of **type I** if it lies between the graphs of two continuous functions of x , that is,

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

where g_1 and g_2 are continuous on $[a, b]$. Some examples of type I regions are shown in [Figure 5](#).

Figure 5

Some type I regions



6

Regions of Type I and II

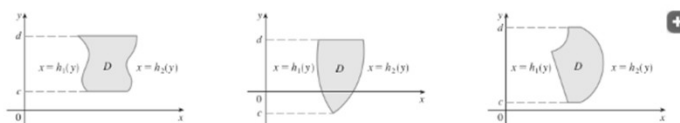
We also consider plane regions of **type II**, which can be expressed as

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

where h_1 and h_2 are continuous. Three such regions are illustrated in [Figure 7](#).

Figure 7

Some type II regions



7

Integrals over Regions of Type I

3 If f is continuous on a type I region D described by

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

then

$$\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$

8

Integrals over Regions of Type II

4 If f is continuous on a type II region D described by

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

then

$$\iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy$$

9

Example: Evaluate the double integral

$$\iint_R (x + 2y) \, dA$$

where R is the region bounded by the parabolas
 $y = 2x^2$ and $y = 1 + x^2$.

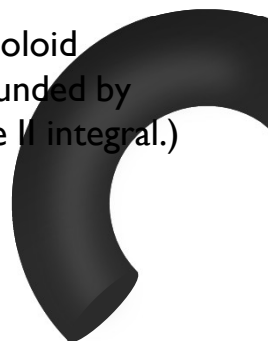
10

Example: Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region in the xy -plane bounded by the line $y = 2x$ and the parabola $y = x^2$. (As a Type I integral.)



11

Example: Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region in the xy -plane bounded by the line $y = 2x$ and the parabola $y = x^2$. (As a Type II integral.)



12

Example: Evaluate the iterated integral

$$\int_0^1 \int_x^1 \sin(y^2) dy dx$$

13

[Extra]

Example: Evaluate the iterated integral

$$\int_0^1 \int_x^1 \sin(y^2) dy dx$$

14

Properties of Double Integrals

$$\iint_D [f(x, y) + g(x, y)] \, dA = \iint_D f(x, y) \, dA + \iint_D g(x, y) \, dA$$

$$\iint_D c f(x, y) \, dA = c \iint_D f(x, y) \, dA \quad \text{where } c \text{ is a constant}$$

If $f(x, y) \geq g(x, y)$ for all (x, y) in D , then

7

$$\iint_D f(x, y) \, dA \geq \iint_D g(x, y) \, dA$$

15

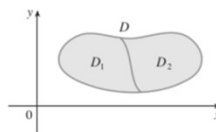
Properties of Double Integrals

If $D = D_1 \cup D_2$, where D_1 and D_2 don't overlap except perhaps on their boundaries (see [Figure 17](#)), then

8

$$\iint_D f(x, y) \, dA = \iint_{D_1} f(x, y) \, dA + \iint_{D_2} f(x, y) \, dA$$

Figure 17



16

Properties of Double Integrals

$$\iint_D 1 \, dA = A(D)$$

10 If $m \leq f(x, y) \leq M$ for all (x, y) in D , then

$$m \cdot A(D) \leq \iint_D f(x, y) \, dA \leq M \cdot A(D)$$

17

[Extra]

Example: Estimate the value of the double integral

$\iint_R e^{-(x^2+y^2)} dA$
 where $R = \{(x, y): x^2 + y^2 \leq 1\}$ is the circle of radius 1.

18

Questions?

19



ALVARO: Start the recording!



20

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Double Integrals in Polar

21

Today – Double Integrals in Polar Coordinates!

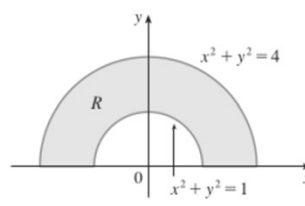
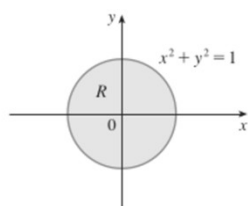
- Polar Coordinates
- Double Integrals in Polar Coordinates
- Change from Cartesian to Polar

22

Polar Coordinates vs Cartesian

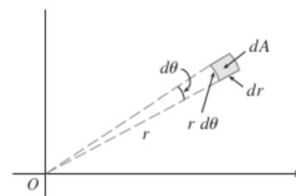
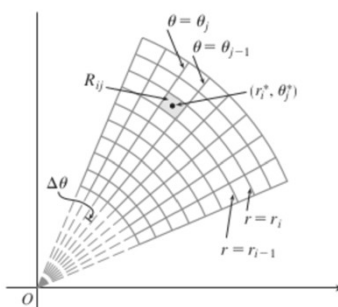
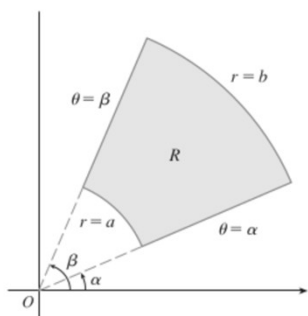
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Example: Express the following regions in polar coordinates.



24

Double Integrals in Polar Coordinates

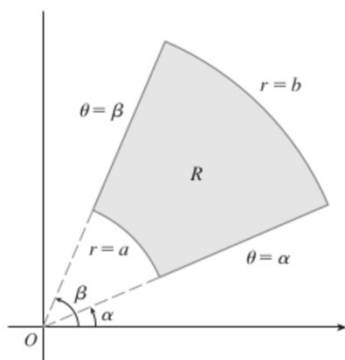


$$\sum_{i=1}^m \sum_{j=1}^n f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) \Delta A_i = \sum_{i=1}^m \sum_{j=1}^n f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) r_i^* \Delta r \Delta \theta$$

25

Double Integrals in Polar Coordinates

If f is continuous on a polar rectangle R given by $0 \leq a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$, then

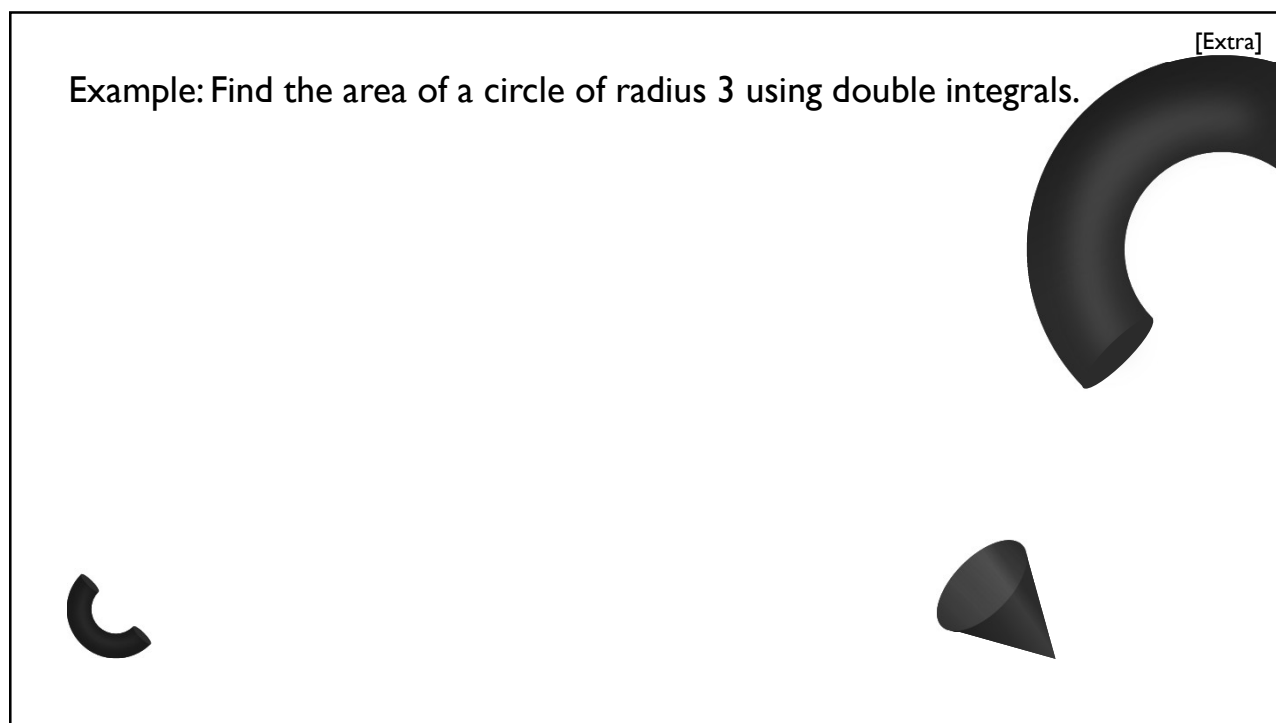


$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$

26

Example: Find the area of a circle of radius 3 using double integrals.

27



28

Example: Find the area of a circle of radius R using double integrals.

29

Example: Evaluate the value of the double integral

$$\iint_R (3x + 4y^2) dA$$

where R is the region in the upper half plane bounded by the circles of radius 1 and 2.

30

[Extra]

Example: Evaluate the value of the double integral

$$\iint_R (3x + 4y^2) dA$$

where R is the region in the upper half plane bounded by the circles of radius 1 and 2.



$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

31

Example: Evaluate the value of the double integral

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx$$



32

[Extra]

Example: Evaluate the value of the double integral

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx$$

33

Example: Find the volume of the solid bounded by $z = 0$ and the paraboloid
 $z = 1 - x^2 - y^2$.

34

[Extra]

Example: Find the volume of the solid bounded by $z = 0$ and the paraboloid $z = 1 - x^2 - y^2$.

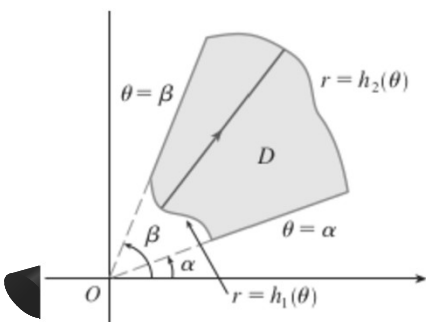
35

Integrals in Polar Coordinates of Type I

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

3 If f is continuous on a polar region of the form

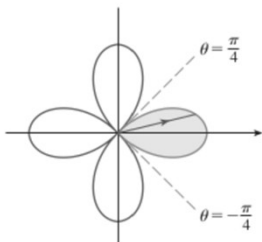
$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$



$$\iint_D f(x, y) \, dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta$$

36

Example: Find the area enclosed by one loop of the four-leaved rose
 $r = \cos(2\theta)$



37

Example: Find the area enclosed by one loop of the four-leaved rose
 $r = \cos(2\theta)$

[Extra]

38

Example: Find the volume of the solid that lies under the paraboloid
 $z = x^2 + y^2$ above the xy -plane and inside the cylinder
 $x^2 + y^2 = 2x$.



39

[Extra]

Example: Find the volume of the solid that lies under the paraboloid
 $z = x^2 + y^2$ above the xy -plane and inside the cylinder
 $x^2 + y^2 = 2x$.



40

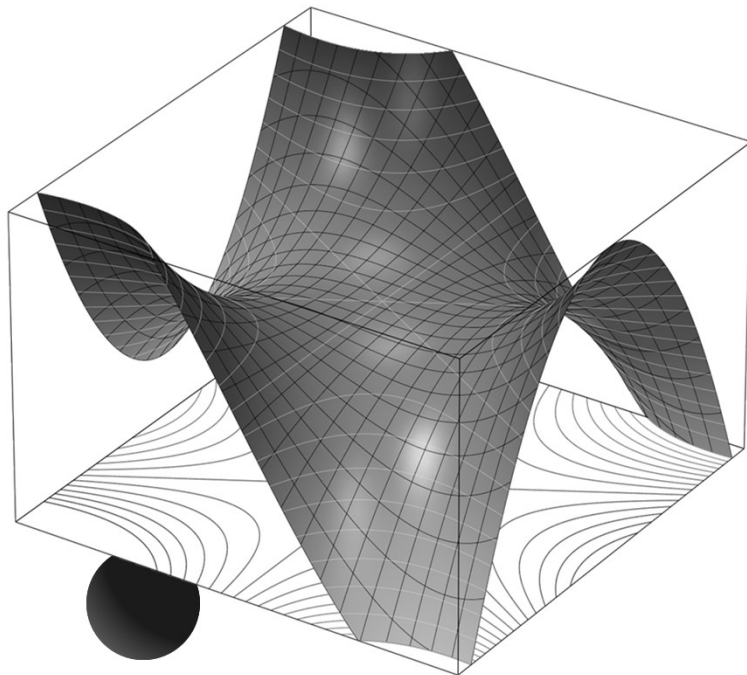
Questions?



41

Thank you

Until next time.



42